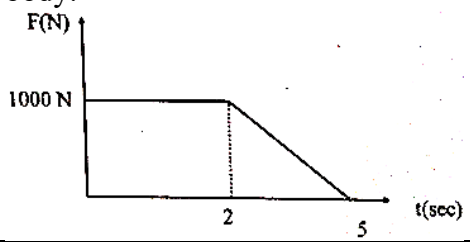
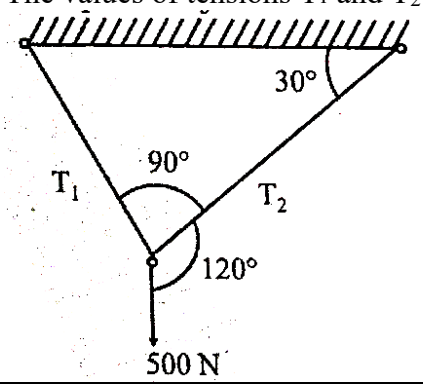
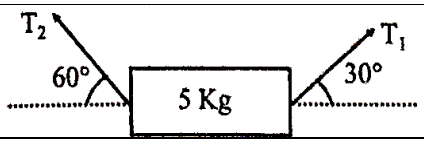
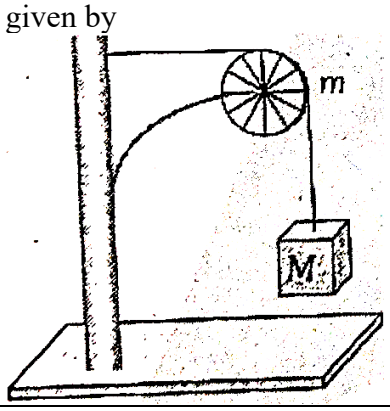
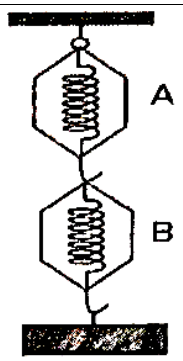
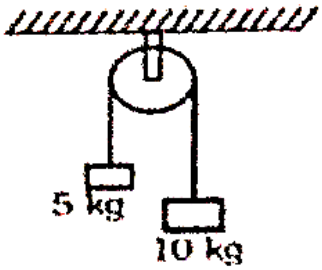
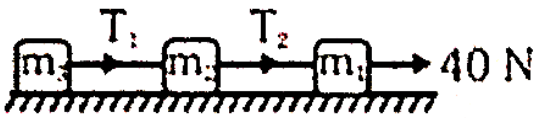
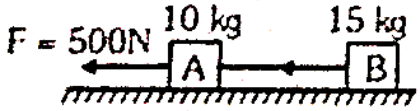
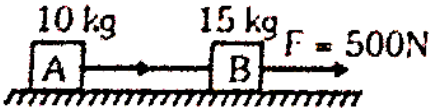
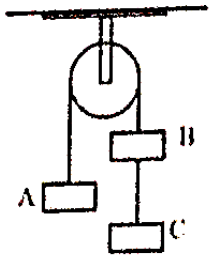
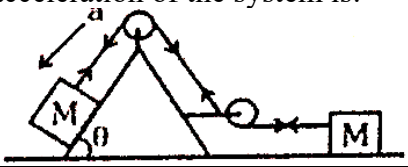
Question	In which of the following direction rock climbers pull the rope to move upwards?
Type	multiple choice
Option	Upwards
Option	Downwards
Option	Sideways
Option	none of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	When a bus suddenly takes a turn, the passengers are thrown outwards because of
Type	multiple choice
Option	Inertia of direction
Option	Acceleration of motion
Option	Speed of motion
Option	Both (b) and (c)
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A force of 50 N acts on a body of mass 5 kg for 2 s. The impulse on the body is
Type	multiple choice
Option	$50\,N-s$
Option	$100\,N-s$
Option	$30\,N-s$
Option	$40\,N-s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Force(F) - time(t) graph is given below. Calculate the impulse experienced by body. 
Type	multiple choice
Option	5000 N-s
Option	3500 N-s
Option	4000 N-s
Option	2500 N-s
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A ball of mass m moves with speed v and it strikes normally with a wall and reflected back normally with v. If its time of contact with wall is t then find force exerted by ball on wall
Type	multiple choice

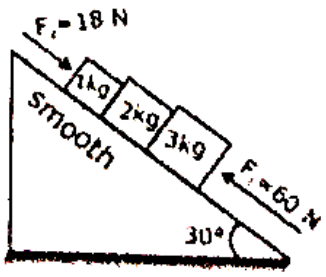
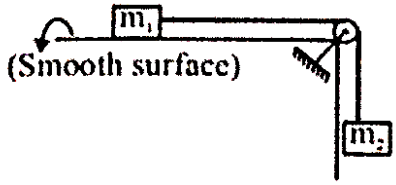
Option	$\frac{2mv}{t}$
Option	$\frac{mv}{t}$
Option	mvt
Option	$\frac{mv}{2t}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A body of mass 5 kg is observed to move with an acceleration of 10 ms^{-2} under the application of a force. The force on the body is
Type	multiple choice
Option	$50\,N$
Option	$5\,N$
Option	$60\,N$
Option	Zero
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is the correct mathematical representation of Newton's 2nd law of motion?
Type	multiple choice
Option	$F = ma$
Option	$F = \frac{mdv}{dt}$
Option	$F = \frac{d(mv)}{dt}$
Option	$F = \frac{md^2x}{dt^2}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Swimming is possible on account of
Type	multiple choice
Option	First law of motion
Option	Third law of motion
Option	Second law of motion
Option	Newton's law of gravitation
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Three concurrent co-planer forces 1N, 2N and 3N acting along different directions on a body
Type	multiple choice
Option	Can keep the body in equilibrium if 2N and 3N act at right angle

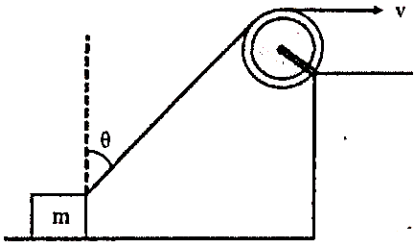
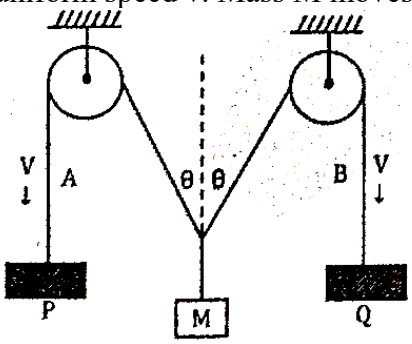
Option	Can keep the body in equilibrium in if 1N and 2N act at right angle
Option	Cannot keep the body in equilibrium
Option	Can keep the body in equilibrium if 1N and 3N act at an acute angle
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A weight of 500 N is supported by two metallic ropes as shown in the figure. The values of tensions T_1 and T_2 are respectively. 
Type	multiple choice
Option	433 N and 250 N
Option	250 N and 433 N
Option	353.5 N and 250 N
Option	250 N and 353.5 N
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Three forces F_1, F_2 and F_3 together keep a body in equilibrium. If $F = 3N$ along he positive $x-$axis, $F_2 = 4N$ along the positive $y-$axis, then the third force F_3 is
Type	multiple choice
Option	5 N making an angle $\theta = \tan^{-1}\left(\frac{3}{4}\right)$ with negative $y-$axis
Option	5 N making an angle $\theta = \tan^{-1}\left(\frac{4}{3}\right)$ with negative $y-$axis
Option	7N making an angle $\theta = \tan^{-1}\left(\frac{3}{4}\right)$ with negative $y-$axis
Option	7N making an angle $\theta = \tan^{-1}\left(\frac{4}{3}\right)$ with negative $y-$axis
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Mayank is doing an experiment in a lab he suspended 5 kg mass by string making angle 60° and 30° with horizontal as shown in figure. Find the tension T_1 and T_2? Take $g = 10 m/s^2$

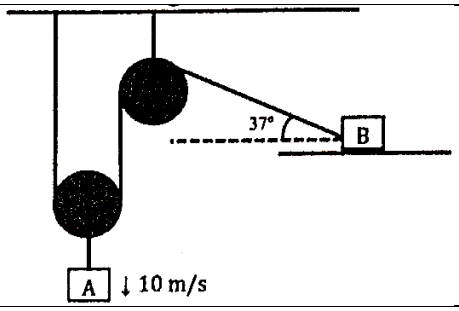
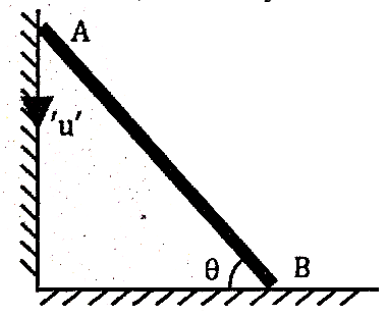
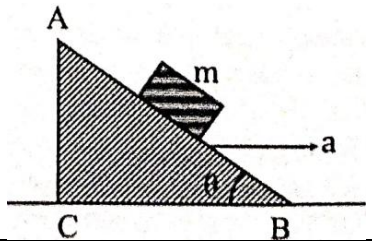
	
Type	multiple choice
Option	$T_1 = 25N : T_2 = 25\sqrt{3}N$
Option	$T_1 = 25\sqrt{3}N : T_2 = 25N$
Option	$T_1 = 25N : T_2 = 25N$
Option	$T_1 = 25\sqrt{3}N : T_2 = 25\sqrt{3}N$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	When a body is subjected to two forces, the body will be in equilibrium if the two forces are:
Type	multiple choice
Option	collinear, equal and act in the same direction
Option	collinear, unequal and opposite
Option	non collinear, equal and opposite
Option	collinear, equal and opposite
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by 
Type	multiple choice
Option	$\sqrt{2}Mg$
Option	$\sqrt{2}mg$
Option	$g\sqrt{(M+m)^2 + m^2}$
Option	$g\sqrt{(M+m)^2 + M^2}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A block of mass 4 kg is suspended through two light spring balances A and B. Then A and B will read respectively.

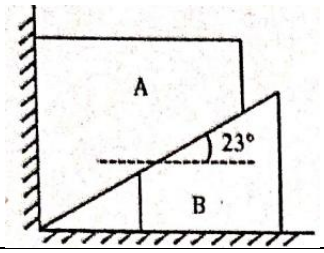
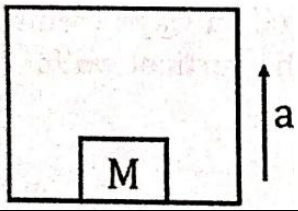
	
Type	multiple choice
Option	4 kg and zero kg
Option	Zero kg and 4 kg
Option	4 kg and 4 kg
Option	2 kg and 2 kg
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Two blocks of masses 5 kg and 10 Kg are connected to a pulley as shown. What will be their acceleration if the pulley is set free? (g = acceleration due to gravity) 
Type	multiple choice
Option	g
Option	$g/2$
Option	$g/3$
Option	$g/4$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Three blocks of masses m_1, m_2 and m_3 are connected by massless strings as shown in the figure on a frictionless table. They are pulled with a force of 40 N. If $m_1 = 10\,kg$, $m_2 = 6kg$ and $m_3 = 4kg$, then tension T_2 will be:- 
Type	multiple choice
Option	$10\,N$
Option	$20\,N$
Option	$32\,N$
Option	$40\,N$
Answer	2
Solution	
Positive Marks	4

Negative Marks	1
Question	Two bodies A and B of masses 10 kg and 15 kg respectively kept on a smooth, horizontal surface are tied to the ends of a light string. If T represents the tension in the string when a horizontal force F = 500 N is applied to A (as shown in figure 1) and T' be the tension when it is applied to B (figure2), then which of the following is true?  Fig. (1)  Fig. (2)
Type	multiple choice
Option	T=T'=500 N
Option	T=T'=250 N
Option	T= 200 N, T'=300 N
Option	T= 300 N, T'=200 N
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Three equal weights each of mass 4kg are hanging on a string passing over a fixed pulley as shown in fig. What is the tension in the string connecting weights B and C. 
Type	multiple choice
Option	Zero
Option	13.3 N
Option	26.6 N
Option	19.6 N
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Two blocks each having a mass M are placed as shown in the figure. The acceleration of the system is:- 
Type	multiple choice
Option	0
Option	$\frac{g\sin\theta}{2}$

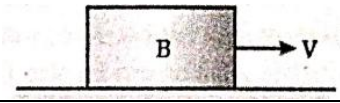
Option	$g\sin\theta$
Option	$2g\sin\theta$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two masses of 10 kg respectively are connected by a massless spring as shown in fig. A force of 200 N acts on the 20 kg mass. At the instant shown the 10 kg mass has acceleration $12m/s^2$ towards right. The acceleration of 20 kg mass at this instant is
Type	multiple choice
Option	$12m/s^2$
Option	$4m/s^2$
Option	$10m/s^2$
Option	Zero
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In the figure $\left(g=10m/s^2\right)$. Acceleration of 2 kg block is: 
Type	multiple choice
Option	$2m/\sec^2$
Option	$4m/\sec^2$
Option	$6m/\sec^2$
Option	$8m/\sec^2$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	 If $m_1=m_2=1kg$ then find out distance travelled by m_1 in first $\frac{1}{2}$ sec.
Type	multiple choice
Option	0.650 m
Option	0.625 m

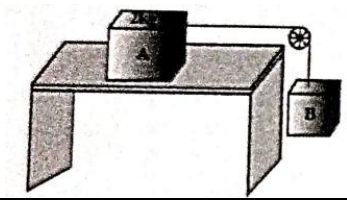
Option	0.725 m
Option	0.525 m
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A block is dragged on a smooth plane with the help of a rope which moves with a velocity v as shown in figure. The horizontal velocity of the block is: 
Type	multiple choice
Option	v
Option	$\frac{v}{\sin\theta}$
Option	$v\sin\theta$
Option	$\frac{v}{\cos\theta}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In the fig., the ends P and Q of an unstrechable string move downward with uniform speed v. Mass M moves upwards with speed. 
Type	multiple choice
Option	$v\cos\theta$
Option	$v/\cos\theta$
Option	$2v\cos\theta$
Option	$2/v\cos\theta$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Find velocity of block B' at the instant shown in figure.

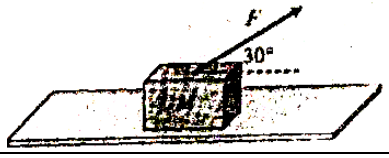
	
Type	multiple choice
Option	25 m/s
Option	20 m/s
Option	22 m/s
Option	30 m/s
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The velocity of end 'A' of rigid rod placed between two smooth vertical walls moves with velocity 'u' along vertical direction. Find out the velocity of end 'B' of that rod, rod always remains in constant with the vertical walls. 
Type	multiple choice
Option	$u \tan 2\theta$
Option	$u \cot 2\theta$
Option	$u \tan \theta$
Option	$2u \tan \theta$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A block of mass m is placed on a smooth inclined wedge ABC of inclination θ as shown in the figure. The wedge is given an acceleration 'a' towards the right. The relation between a and θ for the block to remain stationary on the wedge is: 
Type	multiple choice
Option	$a = \dfrac{g}{\operatorname{cosec}\theta}$
Option	$a = \dfrac{g}{\sin\theta}$

Option	$a = g\ \cos\theta$
Option	$a = g\ \tan\theta$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	There are two blocks A and B in contact with vertical and horizontal smooth surface respectively. Acceleration of A and B are a_A and a_B Respectively along their constrained direction of motion. Relation between a_A and a_B is (Assume $\sin 23^0 = \frac{2}{5}$) 
Type	multiple choice
Option	$2a_A = \sqrt{21}a_B$
Option	$2a_A = 5a_B$
Option	$5a_A = 2a_B$
Option	$\sqrt{21}a_A = 2a_B$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	With what acceleration 'a' should the box of figure moving up so that the block of mass M exerts a force 7Mg/4 on the floor of the box? 
Type	multiple choice
Option	$\frac{g}{4}$
Option	$\frac{g}{2}$
Option	$\frac{3g}{4}$
Option	$4g$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Horizontal force required to move a mass of 1 kg at rest on a horizontal rough plane $\left(\mu = 0.1 \text{ and } g = 9.8 m/s^2\right)$ is
Type	multiple choice

Option	$0.98\ N$
Option	$0.49\ N$
Option	$9.8\ N$
Option	$4.9\ N$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which one of the following statements is incorrect
Type	multiple choice
Option	Rolling friction is smaller than sliding friction
Option	Limiting value of static friction is directly proportional to normal reaction.
Option	Frictional force opposes the relative motion
Option	Coefficient of sliding friction has dimensions of length
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A block of mass 10 kg is placed on a rough horizontal surface having coefficient of friction $\mu = 0.5$. Ifa horizontal force of 100 N is acting on it, then acceleration of the block will be $\left(g = 10\frac{m}{s^2}\right)$
Type	multiple choice
Option	$0.5\ m/s^2$
Option	$5\ m/s^2$
Option	$10\ m/s^2$
Option	$15\ m/s^2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A block of mass 2 kg is kept on the floor. The coefficient of static friction is 0.4. If a force F of 2.5 Newtons is applied on the block as shown in the figure, the frictional force between the block and the floor will be
Type	multiple choice
Option	$2.5\ N$
Option	$5\ N$
Option	$7.84\ N$
Option	$10\ N$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	An iron block of mass 5 kg is kept on a trolley. If the trolley is being pushed with an acceleration of 5 m/s², What will be the force of friction between the block and the trolley surface (Take the coefficient of static friction between the

	block and the surface to be 0.8)
Type	multiple choice
Option	Zero
Option	$5\ N$
Option	$4\ N$
Option	$25\ N$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A block B is pushed momentarily along a horizontal surface with an initial velocity V. If μ is the coefficient of sliding friction between B and the surface, block B will come to rest after a time 
Type	multiple choice
Option	$V/(g\mu)$
Option	$g\mu/V$
Option	g/V
Option	V/g
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A car is moving along a straight horizontal road with a speed v_o. If the coefficient of friction between the tyres and the road is μ, the shortest distance in which the car can be stopped is
Type	multiple choice
Option	$v_0^2/2\mu g$
Option	$v_0^2/\mu g$
Option	$\left(\frac{v_0^2}{\mu g}\right)^2$
Option	v_0/μ
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A body of weight 64 N is pushed with just enough force to start it moving across a horizontal floor and the same force continues to act afterwards. If the coefficients of static and dynamic friction are 0.6 and 0.4 respectively, the acceleration of the body will be
Type	multiple choice
Option	$\frac{g}{6.4}$
Option	$0.64\ g$
Option	$\frac{g}{32}$

Option	$0.2\,g$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A lift is moving downwards with an acceleration equal to acceleration due to gravity. A body of mass m kept on the floor of the lift is pulled horizontally. If the coefficient of friction is μ, then the frictional resistance offered by the body is
Type	multiple choice
Option	mg
Option	μmg
Option	$2\mu mg$
Option	Zero
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A ladder weighing 250 N is placed against a smooth vertical wall having coefficient of friction between it and floor is 0.3, then what is the maximum force of friction available at the point of contact between the ladder and the floor
Type	multiple choice
Option	75 N
Option	50 N
Option	100 N
Option	250 N
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The coefficient of static friction, μ_s, between block A of mass 2 kg and the table as shown in the figure is 0.2. What would be the maximum mass value of block B so that the two blocks do not move? The string and the pulley are assumed to be smooth and massless $\left(g = 10\,m/s^2\right)$. 
Type	multiple choice
Option	$2.0\ kg$
Option	$4.0\ kg$
Option	$0.2\ kg$
Option	$0.4\ kg$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A block of mass M = 5 kg is resting on a rough horizontal surface for which the coefficient of friction is 0.2. When a force $F = 40\ N$ is applied, the acceleration

	of the block will be $\left(g = 10\,ms^2\right)$ 
Type	multiple choice
Option	$5.73\ m/\sec^2$
Option	$8.0\ m/\sec^2$
Option	$3.17\ m/\sec^2$
Option	$10.0\ m/\sec^2$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	An object placed on an inclined plane starts sliding when the angle of incline becomes 30°. The coefficient of static friction between the object and the plane is
Type	multiple choice
Option	$\frac{1}{\sqrt{3}}$
Option	$\sqrt{3}$
Option	$\frac{1}{2}$
Option	$\frac{\sqrt{3}}{2}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A given object takes n times as much time to slide down a 45° rough incline as it takes to slide down a perfectly smooth 45° incline. The coefficient of kinetic friction between the object and the incline is given by
Type	multiple choice
Option	$\left(1-\frac{1}{n^2}\right)$
Option	$\frac{1}{1-n^2}$
Option	$\sqrt{\left(1-\frac{1}{n^2}\right)}$
Option	$\sqrt{\frac{1}{1-n^2}}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The coefficient of friction between a body and the surface of an inclined plane at

	45° is 0.5. If $g = 9.8\ m/s^2$, then acceleration of the body downwards is m/s^2 is
Type	multiple choice
Option	$\frac{4.9}{\sqrt{2}}$
Option	$4.9\sqrt{2}$
Option	$19.6\sqrt{2}$
Option	4.9
Answer	1
Solution	
Positive Marks	4
Negative Marks	1